the correct operation of the system. There are several standard security goals that apply to almost all systems. The most famous of these are the so-called “CIA Triad”:

- *Confidentiality*: ensuring information can only be read by its intended audience.
- *Integrity*: preventing unauthorized creation, modification, or deletion of information.
- *Availability*: ensuring that the legitimate users of an API can access it when they need to and are not prevented from doing so.

Although these three properties are almost always important, there are other security properties that may be just as important in different contexts, such as *accountability* (who did what) or *non-repudiation* (not being able to deny having performed an action). We will discuss security goals in depth as you develop aspects of the sample API.

Security goals can be viewed as *non-functional requirements* (NFRs) and considered alongside other NFRs such as performance or reliability goals. In common with other NFRs, it can be difficult to define exactly when a security goal has been satisfied. It is hard to prove that a security goal is *never* violated because this involves proving a negative, but it’s also difficult to quantify what “good enough” confidentiality is for example.

One approach to making security goals precise is used in cryptography. Here, security goals are considered as a kind of game between an attacker and the system, with the attacker given various powers. The standard game for